A Catalogue of Symmetries and Magic Numbers for Reading Out Spacetime from a State Space

— The Answer-Check Standard of a Research Programme Taking Anonymity and Self-Consistency as Axioms —

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Abstract

The author's research programme asks whether time, space, mass, charge and interactions can be read out from the relations of a set of complex numbers $\{z_n\}$ alone, without placing a background spacetime, a ruler or particle species from outside. The programme has published 156 papers so far, but the numbering of axioms, the symbols and the claims differ slightly from paper to paper, and it had become unclear against what standard a given numerical experiment or thought experiment is to be judged correct.

This paper presents no new derivation. Instead, it fixes a catalogue of the symmetries the programme must satisfy and of the dimensionless numbers (magic numbers) attached to those symmetries, giving every entry a status (established, theory, candidate, open) and a provenance (a published paper or external literature). The catalogue is a reference document that determines which papers later papers should use as their answer-check standard; that is why it contains many self-citations and external citations.

The skeleton of the catalogue is as follows. (1) The zeroth axiom is anonymity and the first axiom is self-consistency. As an axiom derived from anonymity, we place the distinguishability axiom: the readout cannot tell which layer a value belongs to. The squared quantity $z' = z^2$ is also just a complex number, so z and z' cannot be distinguished in the readout. On that basis the closure is defined by a single definition, the linear zero closure of complex numbers $\Sigma z_n = 0$. The square-sum closure $\Sigma z_n^2 = \Sigma z'_n = 0$ is the same definition read at the z' layer, not a new definition. The return of time evolution $U^n = I$ is a theorem. (2) Below these we place the big-picture symmetries (time symmetry of emission and absorption, local and global, the scale of observation, electric–magnetic duality, Galois symmetry, conservation of information), conjugacy and duality, and the projection symmetry. (3) We set out a candidate axiom: an essential symmetry carries a dimensionless magic number rather than the conserved quantity assumed by Noether's theorem. (4) The requirements of the Standard Model, general relativity and E_8, the algebra and geometry of this system, and the locking symmetries of complex systems are listed in the same format. (5) Magic numbers are divided into four systems: integer, coupling-convention, irrational and derived.

We also present two mathematically confirmed results. A linear map that preserves both the sum of true squares and the norm is necessarily a real orthogonal map. The 24 units of the Hurwitz lattice D_4 split into one vector set and two spinor sets (8 + 8 + 8), and triality

cycles them as the quaternion operation $q \mapsto a\,q\,a$ ($a = (-1 + i + j + k)/2$) while preserving the 240 roots of E_8.

All 156 published papers are mapped to the rows of the catalogue: 81 rows have a published standard and 15 rows are set up for the first time in this paper. The axiom numbering is reset in this paper, with a correspondence table to the old numbering. The earlier claim that filled the gap between 137 and α^{-1} with a closed identity is revised to the position that the gap itself is the content to be read. Numerical disagreements, such as the rotation orders of the four-dimensional lattice, the decomposition of 137 and the number of components of one generation, are kept as open problems, with “several interpretations are possible” as the present answer. Finally, we place the root difficulty of the map from the state space to spacetime in the fact that what is visible depends on the projection method, and we show that, since spacetime is stably readable, the map must be at least a partial projection.

Keywords: anonymity, self-consistency, zero square sum, linear zero closure, distinguishability, finite order, symmetry, magic number, Noether’s theorem, conjugacy, duality, partial projection, triality, E_8, mode locking

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1. Position of This Paper

1.1 Why a catalogue is needed

The research programme published 156 papers between April and October 2026. Among the papers that state an axiom system alone, the Anonymous Equal-Amplitude Composite-Wave Model: Basic Axiom System v9.1 [S1], the symmetry generation from zero closure, finite order and self-consistent geometry [S3], the research project design [S14] and the first and second thought experiments [S15, S16] each number their axioms differently (§11). On the side of numerical experiments, no document fixed the criterion for judging whether a result was derived from the skeleton or came from parts imported from outside the skeleton. In the author’s own numerical experiments, a failure occurred in which parts were swapped each time a result did not match until the experiment “passed” (§13.4).

This paper fixes that criterion. Whereas the research project design [S14] fixes the direction of the project (goal image, prohibitions, success conditions), this paper fixes the symmetries and numbers that individual papers and numerical experiments must satisfy. The research project design adopted a policy of no self-citation; this paper, on the contrary, covers self-citations and external citations exhaustively, because it determines which papers are to be used as the answer-check standard.

1.2 How statuses are assigned

Each row of the catalogue is given one of the following four statuses.

Status	Meaning
Established	An established mathematical or physical fact, supported by external literature or by the checks in this paper (§10)
Theory	A proposition set up within the theory of the programme, with a published paper as provenance
Candidate	A conjectured correspondence, not yet checked
Open	Something that stands only as a question, including items admitting several interpretations

1.3 Resetting the numbering

The numbering of this paper (A1–A6, A1′ , Z1, W1–W6, K1–K7, P1, S1–S20, E1–E6, G1–G18, C1–C4, L1–L9) is the standard for later papers. The numbering of published papers is kept as numbering within each paper and is not rewritten. The correspondence with the old numbering is given in §11.1.

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2. Starting Point: The State Is Only a Set of Complex Numbers

The state of this system is only a set of complex numbers

$$\{z_n\}, \qquad z_n = a_n + i\, b_n.$$

In a sense it is a Hilbert space, and there are no positions, lattice points, values sitting on vertices, fields or background spacetime. Each z_n is merely a point in the complex plane, and its magnitude

$$R^2 = a^2 + b^2$$

is read not as the radius of some space but as the magnitude of the complex number itself.

Spacetime {xyzt}, distance, time and R are all read out from the relations among the z_n (ratios, phase differences, true squares, cross terms). The largest misreading the author repeated during the research was to believe that the background is xyz space. Under that misreading one regards lattices and spheres as containers and tries to arrange complex numbers in them. In this paper, lattices, S^3 and Clifford translations are treated as structures in phase space, not as arrangements in xyz space.

The destination of the readout is split into the eight directions of two quaternions:

$$A = t + xi + yj + zk \quad (\text{readable side}), \qquad B = R + Q_1 i + Q_2 j + Q_3 k \quad (\text{unreadable interior}).$$

The dimensions of number systems that preserve magnitude under squaring are limited to 1, 2, 4 and 8 (Hurwitz's theorem), and eight directions are the upper limit. Setting R and $Q_1Q_2Q_3$ up as independent axes avoids the confusion by which Weyl's 1918 scale gauge theory made R (scale) also carry the role of Q (charge) and reached an impasse. t is the clock of space and R is the internal clock (mass), so two clocks can be read separately.

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3. Axioms and Theorems

3.0 The distinguishability axiom and the single definition of closure

No.	Content	Kind
A1′	**Distinguishability axiom**: the readout cannot tell which layer a read-out value belongs to. This applies anonymity (A1) to layers rather than to elements	Axiom (derived from A1)

No.	Content	Kind
—	**Its appearance in the readout**: the squared quantity $z' = z^2$ is also just a complex number, so z and z' cannot be distinguished in the readout	Symmetry
—	**Choice of basic quantity**: the squared quantity $z'_n = z_n^2$ is also a basic quantity. Writing $z' = A + iB$, $A = a^2 - b^2$ and $B = 2ab$	Definition
Z1	**Linear zero closure**: $\Sigma z_n = 0$ ($\Sigma a_n = 0$ and $\Sigma b_n = 0$). The single definition of closure in this paper	Definition
—	**Its consequence**: the square-sum closure $\Sigma z_n^2 = \Sigma z'_n = 0$ is Z1 read at the z' layer, not a new definition	Consequence

Under the distinguishability axiom, the one definition “the sum of the readable values is 0” is imposed so that it holds whichever layer it is read at. As a result, both $\Sigma z_n = 0$ and $\Sigma z_n^2 = 0$ follow. What looked like two conditions is one definition seen in two readings. Together they give

$$\sum_n a_n = \sum_n b_n = 0, \qquad \sum_n a_n^2 = \sum_n b_n^2, \qquad \sum_n a_n b_n = 0,$$

and $(\Sigma a_n)(\Sigma b_n) = 0$ follows trivially. If the points (a_n, b_n) are placed in the complex plane, their centroid is at the origin, their spreads along the two axes are equal, and the two axes are uncorrelated. Up to second order, the point set has no special origin and no special direction, and no meaning is left in which axis is called real and which imaginary. This is the first- and second-order expression, on the complex plane, of Axiom 0 of the basic axiom system (no privileged axis, sign or type) [S1].

The linear sum is the lowest-degree quantity among those unchanged by every permutation, so the definition of closure can be written with only the lowest-degree invariant of anonymity and the indistinguishability of layers. The squaring map $z \mapsto z^2$ is not part of the definition; it stands outside the definition as the map connecting the two readings.

Provenance: a precedent that used the linear closure as a condition on parent states (centroid at the origin) is the complex-simplex study of 2026-08-30. The view that places the linear closure as the definition and identifies it with the square-sum closure, and the distinguishability axiom, come from the thought-experiment dialogue record on which this paper is based (Appendix D).

3.1 Two axioms

No.	Axiom	Content	Forbids
A1	**Anonymity** (zeroth axiom)	Elements carry no labels; there is no special origin, coordinate, distance, sign or type. Only relations unchanged by every permutation of the elements are readable	Placing a background by hand, distinguishable vertices, a ruler given from outside, an observer standing outside
A2	**Self-consistency** (first axiom)	There is no external reference, so the system takes itself as its reference. The loop in which the state determines itself and that state again determines itself closes	Placing the observer or R outside the system

Provenance: A1 is Axiom 0 of the basic axiom system [S1]; A2 is A4 of the zero-closure paper [S3].

3.2 Closure and return

No.	Content	Explanation	Status
A3	**Closure of true squares** $\Sigma z_n^2 = 0$ (Z1 read at the z′ layer)	When the whole closes consistently with itself without an external scale, the sum of true squares (squares without conjugation) closes. The 0 is not a value but a statement that a conserved quantity can exist. By §3.0 it is not an independent definition but the reading of the linear closure Z1 at the layer of squared quantities	Theory. A proof as a theorem following from self-consistency is in [S12]; that nontrivial closure requires complex numbers is in [S11]

No.	Content	Explanation	Status
A4	**Return** $U^n = I$	There is no external clock, so time evolution can only be written as a permutation among indistinguishable elements. A permutation of finitely many indistinguishable elements has finite order and always returns somewhere	Theory. Finite-order resonance in [S7], selection of roots by finite order in [S20], the exact solution of the Z_4 phase in [S23], the minimal exact example in [S24]

A3 and A4 correspond to the two ways of closing. A3 is the closure of structure, on the side where the three dimensions of space support stationary states while remaining continuous. A4 is the closure of time, on the side where four-dimensional spacetime becomes discrete (§3.4). The fourteenth thought experiment [S19] treated $\Sigma x^2 = 0$ and $U^n = I$ not as assumptions placed at the start but as a common form appearing when structures of different character are combined. This paper follows that discipline and does not take either as a starting assumption: A3 is a reading of the definition Z1 of §3.0, and A4 is a theorem.

3.3 The closure equation does not fix the scale

With $z_n = a_n + i\, b_n$,

$$\sum_n z_n^2 = \sum_n (a_n^2 - b_n^2) + 2i \sum_n a_n b_n,$$

so $\Sigma z_n^2 = 0$ requires both $\Sigma a_n^2 = \Sigma b_n^2$ and $\Sigma a_n b_n = 0$. Both are homogeneous equations and hold when every z_n is multiplied by the same complex number λ ($\Sigma(\lambda z_n)^2 = \lambda^2 \Sigma z_n^2$). Scale invariance and invariance under the global phase are built into the closure equation from the start. The imaginary part of the closure, $\Sigma a_n b_n = 0$, is also the condition that the generator of scale transformations vanishes [S1, S6].

Therefore the closure does not fix the unit of the ruler. What fixes the ruler remains a separate question (§13).

3.4 Three dimensions are continuous, four dimensions are discrete

- **Three-dimensional space can form self-consistent states while remaining continuous.** Only on odd-dimensional spheres do waves propagate without tails and refocus sharply at the antipode (Huygens' principle). The spheres on which uniform translations form a group are only S^1 and S^3. Circular waves and localized waves close as standing waves in space and need not advance their phase.
- **Introducing time makes the phase rotate.** For self-consistency to keep closing, the rotation of the phase must return exactly (A4). With continuous time and a generic angle, the phase never returns to the same place and drifts quasi-periodically. To keep closing, the rotation angle must be a rational fraction, and time becomes discrete.
- **Four-dimensional spacetime must be discrete.** For spatial translation and phase rotation to be compatible within one structure a lattice is needed, and the rotation orders compatible with a lattice are restricted to 1, 2, 3, 4, 6 by the crystallographic restriction. Four-fold and six-fold symmetries coexist on the same lattice in the Hurwitz lattice D_4 (24 units). There are six regular polytopes in four dimensions, more than the three in five or more dimensions.

3.5 Other theorems and propositions

No.	Content	Status	Provenance
A5	**0 + 0 = 0 (composition)**: the closure of parts becomes the closure of the whole. Decomposition is hard; composition is easy	Theory	[S25]
A6	**Compatibility of the norm and the true square**: a linear map preserving both is necessarily real orthogonal	Established (§10.1)	[S22], Check 1 of this paper

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4. The Layer Just Below the Axioms: Big Picture, Conjugacy and Duality, Numbers, Projection

4.1 Big-picture symmetries

These are symmetries concerning the state of the whole system rather than individual numbers or groups. Before entering individual symmetries, this layer is checked first.

No.	Symmetry	Related to	Conserved quantity / fixed point	Status
W1	Time symmetry of emission and absorption, and the arrow of time	The perfect absorber at infinity. The boundary condition of the absorber breaks an equation that is time-symmetric in retarded and advanced waves and selects the arrow of time (Wheeler–Feynman [X27])	The time-symmetric pair of fields. The record on the absorber increases monotonically	Candidate. The absorber map is in [S17]

No.	Symmetry	Related to	Conserved quantity / fixed point	Status
W2	Local and global (Mach's principle, cosmological principle)	Local R is determined by global R_0. Homogeneity and isotropy of the universe	The representative curvature R_0 made by the eigen-oscillation of the whole, which cancels local curvature and closes to zero	Candidate
W3	The scale of observation (renormaliza-tion group), micro and macro	α moves continuously between 128 and 137 with the energy of observation. A closed system owns its scale. Dirac's large numbers, the resolution N_0 of the absorber	Renormaliza-tion fixed points, scale-independent ratios	Candidate
W4	Electric–magnetic duality	The Hopf fibration is the bundle of the Dirac monopole. A single monopole quantizes charge	Electric and magnetic charge (topological)	Candidate (set up for the first time here)
W5	Symmetry on the side of numbers (Galois symmetry)	Which rationals are locked onto and which irrationals remain. The projection that makes the 600-cell quasicrystal from E_8 uses the Galois conjugation sending the golden ratio φ to $-1/\varphi$	Quantities unchanged by conjugation (field norm and trace)	Candidate
W6	Conservation of information	In a closed system information is not lost and is recorded in the gap at the boundary	The total amount of record (boundary area, resolution N_0)	Candidate. The minimal unit of information is in [S13]

4.2 Conjugacy and duality

Conjugacy. Physical quantities always appear as pairs with a fixed product: frequency and wavelength ($\nu\lambda = c$), position and momentum, time and energy, mass and Compton wavelength, near and far (large and small view angles), large and small (R and 1/R). The question at which the programme was long stuck was what the conjugate of the amplitude A is. As a candidate this paper names the pair of amplitude and phase. In the exponential representation $E = \ln A + i\varphi$, $\ln A$ and φ pair up as the real and imaginary parts of the same complex number E. In quantum optics the number N ($\propto A^2$) and the phase are conjugate. If the phase is periodic and becomes discrete by $U^n = I$, its conjugate (the number) is quantized to integers (duality of the circle U(1) and the integers $\mathbb{Z}$). This is a candidate for the origin of the unit of the ruler (status: candidate). Precedents for conjugate pairs are $\nu\lambda = 1$ in the dual-geometry series [S8] and the discussion of Pontryagin duality [S10].

Duality. Duality is the structural version of conjugacy. There are two sides, such as inside and outside, spacetime and interior, front and back; the same conserved quantity exists on both sides, but it is called by a different name on each side. In Kaluza–Klein theory the momentum along the fifth axis is called charge; in $E = mc^2$ the same thing is called energy and mass; in holography the area and entropy of the boundary are called the energy of the interior. Within this system too, the clock of space t and the internal clock R divide one rotation by $v^2 + (1/\gamma)^2 = 1$, and it is called velocity on one side and mass on the other.

Combination with anonymity. Under anonymity, the two sides of a duality carry no labels. Which side one reads from is only a choice of labels, and conserved quantities called by different names on different sides are one and the same quantity. Hence different symmetries, conserved quantities and fixed points are often different views of one symmetry. In the duality schema of De Haro and Butterfield [X12], the “common core” shared by two models is the content of the theory, and the interpretation maps assign meanings. In the language of this paper, the common core corresponds to the same conserved quantity whose name changes, and the interpretation map to the renaming. If renamings become indistinguishable under anonymity, dual models become the same thing. This is the claim this paper adds.

Seven groups that may merge into one are listed as candidates.

Group	Rows that may merge	Common content
K1 Allocation of one rotation	A3, S3 (Lorentz), S4 (signature), S9 (cross term 2ab), the two clocks t and R	Invariant mass = rest frequency = rest energy
K2 Return of one step	A4, S2 (energy), G8 (crystallographic orders), L3 (orders 4 and 6 of the modular group), L6 (FPUT recurrence), G14 (parity), G18 (order of the exchange)	Rotation of finite order
K3 Winding	G5 (winding number of the Clifford translation), S1 (momentum), G13 (Hopf winding number), W4 (magnetic charge), S6 (U(1)), Kaluza–Klein momentum	Integer winding number
K4 Inside and outside	G16, W6, W1, S5, W2	Record on the boundary = energy of the interior
K5 Oneself as a fixed point	A2, L4 (renormalization fixed point), W3, the scale freedom of A3, amplitude and phase	A point returning to itself when the scale is changed

Group	Rows that may merge	Common content
K6 Permutation of three	G11 (triality), S17 (spin and statistics), S19 (three generations), E2, S8	Permutation of a vector and two spinors
K7 Labels of sides	A1, S10–S13 (C, P, T, CPT)	Which charge is positive, which side is left, which direction is past are choices of labels. The CPT theorem, invariance under changing all three labels at once, is a candidate expression of the anonymity of sides

Whether a group merges is checked by whether the conserved quantities of the group agree up to renaming alone.

4.3 Dimensionless numbers attached to symmetries (candidate axiom of magic numbers)

Noether's theorem says that each continuous symmetry has a conserved quantity. The author does not regard this as an essential axiom. The candidate axiom set up in this paper is:

> An essential symmetry carries not a conserved quantity but always a dimensionless magic number. These are the numbers hidden behind e, π and α, and they are fixed almost entirely by number theory, group theory and geometric structure.

This system is scale invariant, and only ratios are physical (§3.3). What a symmetry fixes is not a dimensional conserved quantity but a dimensionless number. A conserved quantity is only that number read out after multiplying by a ruler. In a world of ratios only, every ratio is preserved however the ruler is changed, and there are no special values. What does not vanish under a change of scale is only the pure numbers determined by structure. Integer numbers (orders, winding numbers, counts) select allowed values discretely from continuous ratios and bring in discreteness. Numbers that determine groups, lattices and geometries (π, φ, E_8's 240 and 248) determine the form of relations and bring in structure. Scale invariance cannot break itself, so in a system that brings in no ruler from outside, the only source of discreteness and structure left is magic numbers.

Known examples abound. The nuclear magic numbers 2, 8, 20, 28, 50, 82, 126 come from adding a spin–orbit term to the SU(3) symmetry of the three-dimensional harmonic oscillator (2, 8, 20, 40, 70, 112) [X30]. Universal constants at renormalization fixed points (the Feigenbaum constants, the dimension of the devil's staircase) are determined by structure alone, independently of the details of the system [X23]. E_8 has many numbers, in proportion to the many symmetries imposing constraints (§9).

When different symmetries require nearby numbers, the observed value does not settle on any one of them; it is pulled in and oscillates metastably between them. Observed constants are close to, but not equal to, structural numbers. Eddington demanded agreement by claiming that α^{-1} is the integer 136, later 137, while the mainstream discarded the structural numbers themselves because of the gap [X18]. This paper takes neither side and treats the gap not as an error but as content to be read. This relation is why a world that is uncertain and discrete looks continuous and deterministic. Because structural numbers give the points of locking, jumps are observed discretely. Because several numbers compete, observed values drift continuously with scale (the running of α). Quasi-periodic motion that is not locked takes exponentially long to break down (KAM and Nekhoroshev), and meanwhile it looks like a fixed orbit.

4.4 The projection symmetry

No.	Symmetry	Related to	Conserved quantity / fixed point	Status
P1	Full, partial and zero projection of an operation interchange with the projection method	The map between state space and spacetime, uncertainty, discreteness, the Born rule, the exchange ratio	The ratio of projection (a fixed point of locking if stable)	Hypothesis. Axiom 0.7 of the basic axiom system [S1], the anonymous partial projection paper [S2]

The same state and the same operation can look like a full projection, a partial projection or a zero projection depending on the projection through which they are viewed. A complex number looks like a full projection in one projection (it maps onto all values) and like a zero projection in another (multiplying by i leaves zero projection onto the original direction). The two are not contradictory; they are different projections.

How it looks	Examples	What comes out
As a full projection	The norm (always positive, never 0), magnification and reduction	Continuity, definiteness, determinism
As a partial projection	Rotation by an angle θ (cos θ remains in the original direction), measurement of a superposition (probability $\cos^2\theta$), the right exchange R = $\cos^2\theta$	Probability, locking ratios
As a zero projection	Multiplication by i, the antisymmetric part of a product (commutator), measurement of an orthogonal state	Uncertainty (bounded below by the commutator, [x, p] = iℏ), discreteness ($i^4 = 1$)

The map from the state space $\{z_n\}$ to spacetime corresponds to viewing the state space through some projection. Until the projection method is chosen, what is readable and what is not is undetermined. **This is the root difficulty of the map between state space and spacetime.** Since spacetime is relatively stably readable, the projection must be at least a partial projection. With a zero projection spacetime would not appear; with a full projection neither the unreadable interior ($RQ_1Q_2Q_3$) nor uncertainty nor discreteness would remain. With a partial projection, the projected part becomes readable spacetime {xyzt}, the part escaping orthogonally becomes the unreadable interior and phase, and uncertainty and discreteness come from the escaped part. For spacetime to be stably readable the ratio of projection must also be stable, and a stable ratio should be locked onto some structural number. The right exchange R = $\cos^2\theta \approx 0.7$ has the form of a partial-projection ratio, but whether it is the candidate is unknown (candidate).

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5. Symmetries Required by the Standard Model, General Relativity and E_8

External standards are listed with their candidate correspondences in this system.

No.	Symmetry	Conserved quantity	Noether's theorem	Correspondence in this system	Status
S1	Spatial translation	Momentum	Holds; not localizable in gravity	Left Clifford translation (fixed-point free; winding number and one-step angle are gauge invariants)	Candidate
S2	Time translation	Energy	Holds classically	Eigenphase of $U^n = I$	Candidate
S3	Lorentz transformation	Angular momentum, boost charges	Holds	Decomposition $z^2 = (a^2 - b^2) + i{\cdot}2ab$, $v^2 + (1/\gamma)^2 = 1$, additivity of rapidity	Identity is theory [S16, S21]
S4	Signature of spacetime	—	—	Relocating terms by multiplying a coordinate by i; groups of different signature all lie in the same $O(n, \mathbb{C})$	Established (§10.1)
S5	General coordinate transformation, equivalence principle	No local conserved quantity (identity by the second theorem)	Doubtful	Stretch γ by a localized wave, deficit angles of view-angle triangles	Candidate [S15]
S6	U(1)_Y and U(1)_EM	Charge	Holds	Hopf winding number, the charge formula of sign vectors $Q_{em} = -(1/6)(Q_1 + Q_2 + Q_3) - t/2$	Candidate
S7	SU(2)_L	Weak isospin	Spontaneously broken	One of SU(2)_L × SU(2)_R, simultaneous flip of (t, R)	Candidate
S8	SU(3)	Colour charge	Holds	$G_2 \supset SU(3)$ fixed by triality, or the 3 + 2 decomposition [S3, S18]	Candidate

No.	Symmetry	Conserved quantity	Noether's theorem	Correspondence in this system	Status
S9	Electroweak symmetry breaking	—	Broken	Pinned cross term 2ab (spin 0)	Candidate
S10–S13	C, P, T, CPT	—	Discrete	Sign flip of Q, quaternion conjugation $q \mapsto \bar{q}$ (exchange of left and right multiplication), complex conjugation	Candidate
S14	CP violation	—	Broken	Mismatch between the exchange rotation and the lattice	Candidate
S15	Baryon number, lepton number	B, L	B + L anomalous at the quantum level	Components of sign vectors	Open
S16	Chiral symmetry	Axial charge	Broken by quantum anomaly	Left and right multiplication [S4]	Candidate
S17	Spin and statistics	Type of statistics	Follows from Lorentz invariance	Parity $(-1)^n$	Candidate
S18	Gauge anomaly cancellation	Sum of charges per generation is 0	Consistency condition	Charges of sign vectors come out without excess or deficit [S4]	Candidate
S19	Three generations	—	Reason unknown	Triality (order 3)	Candidate
S20	Scale and conformal symmetry	Dilation charge	Broken by mass and quantum anomaly	$\Sigma z^2 = 0$ is trivially scale invariant	Theory (§3.3)

No.	Where E_8 appears	Correspondence in this system	Status
E1	Gluing of the readable side and the interior	Pairing two D_4 lattices in the same coset gives the 240 roots of E_8	Established (§10.2); precedent [S2]
E2	$E_8 \supset SO(16) \supset D_4 \times D_4$	Triality of the pair preserves the roots of E_8	Established (§10.2)

No.	Where E_8 appears	Correspondence in this system	Status
E3	Grand-unification chain $E_8 \supset E_6 \supset SO(10) \supset SU(5)$	—	Open
E4	The spinor 16 of SO(10)	Same number as the 16 even sign vectors $tRQ_1Q_2Q_3$	Candidate (set up for the first time here)
E5	Heterotic string, Lisi's E_8 theory	—	Reference
E6	Contact between E_8 and 137	—	Open

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6. Symmetries of the Algebra and Geometry of This System

No.	Symmetry	Related to	Status	Provenance
G1	Composition algebras (dimensions 1, 2, 4, 8)	The eight directions	Established	Hurwitz's theorem
G2	$S^3 = SU(2)$, uniform translations form a group	The three dimensions of space (only S^1 and S^3; S^7 violates associativity)	Established	Adams' theorem [S21]
G3	Huygens' principle	Stationarity of localized waves	Established	[S21]
G4	Mass gap from curvature	The -1 in the S^3 eigenvalues $(n + 1)^2 - 1$	Established	[S21]
G5	Left multiplication (Clifford translation)	An equally spaced scale unaffected by the distortion of space (gauge)	Theory	[S21]
G6	Right multiplication (array of component rotations, exchange)	Interaction, $R = \cos^2\theta \approx 0.7$	Theory	[S7, S27]
G7	Continuous rotations in three dimensions	Angular momentum; the centre θ_0 is a fixed point	Established	—
G8	Crystallo-graphic symmetry in four dimensions	Discreteness of spacetime; orders 1, 2, 3, 4, 6 (see the open item in §11.3)	Theory	§3.4

No.	Symmetry	Related to	Status	Provenance
G9	The 24 units of the R = 3 system	Multiplication from left or right preserves the lattice H and D_4; the ball centre (½,½,½,½) is one of the units	Established	Checks of this paper
G10	Odd R	Vertices lie on a cell-centred hypersphere only for odd R; R = 3 gives 137	Established (count)	[S9, S31, S32]
G11	Triality	One vector set and two spinor sets, candidate for three generations	Established (math), candidate (physics)	§10.2, [X28]
G12	Quaternion form of triality	$q \mapsto a\,q\,a$	Established	§10.2 (set up for the first time here)
G13	Hopf fibration	Charge (winding number)	Theory	[S21]
G14	Parity $(-1)^n$, antipode	Fermions and bosons	Theory	[S21]
G15	Sign vectors $tRQ_1Q_2Q_3$	16 even ones form a generation; xyz carries spin	Theory	This paper (study of 2026-10-16)
G16	Duality of inside and outside	Cross-polytope and hypercube, particle and absorber	Candidate	The mathematics of the cross-polytope–hypercube duality is in [S28]
G17	Central projection	R is unreadable; the origin is shared	Theory	Axiom 3 of the basic axiom system [S1], chapter 11 of the fourteenth [S19]
G18	Exchange rotation 23π/124	Finite order, not crystallo-graphic; not a symmetry of the lattice	Established (not a lattice symmetry)	[S7, S19]

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7. Symmetries of Locking and Metastable Oscillation

When nearby discrete values are required by different symmetries, the system may not lock onto any of them and may instead oscillate metastably or asymptotically with an extremely long period. Symmetries found by complex systems and nonlinear dynamics are listed with

their correspondences in this system.

No.	Known structure	Content	Correspondence in this system	Status
L1	Arnold tongues, circle map	Frequency ratios lock onto nearby rationals; the width narrows as the denominator grows	Locking between the exchange rotation and the lattice orders; measured in the model in [S26]	Established (math), candidate (correspondence)
L2	Devil's staircase	At criticality a step stands at every rational; the remainder is a Cantor set of dimension about 0.868 [X23]	The structure, as a set, of being near but not equal	Established, candidate
L3	Farey tree, modular group SL(2, $\mathbb{Z}$)	Steps are ordered by mediants $(p + p')/(q + q')$ [X24]. The modular group is generated by elements of order 4 and 6	The crystallographic orders Z_4 and Z_6 enter the symmetry of the locking hierarchy. In the quantum Hall effect the subgroup $\Gamma_0(2)$ gives the selection rule of transitions [X25]	Established, candidate
L4	Golden ratio, renormalization fixed point	The ratio least prone to locking; scale symmetry of self-similarity	Limit of metastability	Established, candidate
L5	KAM theorem, Nekhoroshev stability	Quasi-periodicity survives; even when it breaks, the time to break down is exponentially long	Asymptotic oscillation with an extremely long period	Established, candidate
L6	FPUT recurrence	No approach to thermal equilibrium; near return after a long time	Approximate return of $U^n = I$	Established, candidate

No.	Known structure	Content	Correspondence in this system	Status
L7	Kuramoto model	Transition between synchrony and asynchrony	How the phases of many z_n align; the single-wave equation is of Kuramoto type [S22]	Established, candidate
L8	Quasicrystals	Projecting E_8 to four dimensions gives a quasicrystal with the 600-cell symmetry H_4, with the golden ratio entering [X26]	A symmetry incompatible with the lattice appears as an order that is near but not equal	Established, candidate
L9	Theta function of E_8	A modular form (Eisenstein series of weight 4)	Contact between E_8 and the modular group	Established, candidate

Locking appears not as a conserved quantity of a flow but as a fixed point (the lock) and quasi-oscillation around it. The symmetries of the devil's staircase, the Farey tree and renormalization fixed points are discrete hierarchies and scale self-similarity, outside Noether's theorem.

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8. Where Noether's Theorem Fails or Is Doubtful

1. **Discrete symmetries** (A1, A4, G8, G11, G14, G15): there is no current. What is conserved is an eigenvalue (a phase or a sign).
2. **Fixed points** (A2): self-consistency is a fixed point rather than a symmetry; the conserved quantity and the fixed point are the same thing.
3. **Topological conservation** (G13, W4): winding numbers are conserved by topological structure rather than by symmetry.
4. **Scale** (A3, S20): there is a symmetry, yet the closure does not fix the ruler.
5. **Gravity** (S5): local stretching breaks translation symmetry and energy cannot be localized. By Noether's second theorem conservation laws become identities holding identically ("strong laws") [X8].
6. **Quantum anomalies** (S15, S16, S20): the axial U(1), B + L and scale symmetry are conserved classically but broken by quantization [X9].
7. **Big-picture symmetries** (W1, W4, W5): the arrow of time is broken by a boundary condition, and electric–magnetic duality and Galois symmetry lie outside Noether's theorem.
8. **Mismatch of the exchange** (G18): there is no exact conserved quantity, only locking and quasi-oscillation.
9. **Locking symmetries** (§7): they appear as fixed points and quasi-oscillation around them.

All of these lie on the side of the view of §4.3 that what is attached to a symmetry is not a conserved quantity but a number.

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9. Catalogue of Magic Numbers

Numbers that look like magic numbers are divided into four systems by their origin. When a new number appears, first decide its system. Mistaking a derived number for an integer or irrational one leads into the trap of demanding agreement.

9.1 Integer system (numbers required by structure)

Easily overlooked integers: 0 (closure, a statement of existence), 1 (one closure, $v^2 + (1/\gamma)^2 = 1$), 2 (one complex number cannot close; the minimal system is a two-body system), 3 (the dimension of space, triality, three generations), 4 (the dimension of spacetime, quaternions), 8 (the upper limit of composition algebras, the eight directions), n (the order of $U^n = I$).

Integers required as sequences (mostly from number theory, group theory and geometry):

Sequence	Origin	Status
1, 2, 4, 8	Dimensions of composition algebras	Established
1, 3, 7	Dimensions of parallelizable spheres (S^1 and S^3 form groups)	Established
1, 3, 7, 15	Dimensions of the total spaces of Hopf fibrations	Established
1, 2, 3, 4, 6	Rotation orders compatible with a lattice	Established
5, 6, 3	Numbers of regular polyhedra and polytopes (three, four, five or more dimensions)	Established
2, 6, 12, 24, 240, 196560	Kissing numbers (dimensions 1, 2, 3, 4, 8, 24)	Established
24, 48, 120	Orders of the binary tetrahedral, octahedral and icosahedral groups	Established
14, 52, 78, 133, 248	Dimensions of the exceptional Lie groups G_2, F_4, E_6, E_7, E_8	Established
2, 8, 20, 28, 50, 82, 126	Nuclear magic numbers	Established
1, 137, 1545, 7281	Numbers of hypercubes inscribed in a cell-centred hypersphere (R = 1, 3, 5, 7)	Established (count) [S9]
1, 2, 3, 7, 11, 19, 43, 67, 163	Heegner numbers	Established
8	Bott periodicity	Established

Integers of E_8 and related lattices ([X29]): rank 8, dimension 248, 240 roots, Coxeter number 30 (also 30 for the 600-cell group H_4), Weyl group order 696,729,600, determinant 1. The exponents of E_8, 1, 7, 11, 13, 17, 19, 23, 29, are the integers below 30 coprime to 30; their number is $\varphi(30) = 8$ = rank, and their sum 120 is the number of positive roots, equal to the number of vertices of the 600-cell. The binary icosahedral group (the 120 vertices of the 600-cell) corresponds to E_8 by the McKay correspondence. For D_4: 24 roots, Weyl group order 192, 1152 automorphisms (§10.2).

Ratios of integers: 1/2 of spin and of Dirac's condition, charges 1/3 and 2/3, 16 per generation, 12 gauge particles (8 + 3 + 1).

9.2 Coupling-convention system (numbers set by unit conventions)

G = c = q_0 = 1, q_0 = e/3, J = ħ/2, natural units ħ = c = 1. These are not magic numbers but numbers set to 1 or 3 by the choice of units. Only dimensional constants can be set to 1 by a unit convention; dimensionless numbers (α and others) cannot be changed by the choice of units.

9.3 Irrational system

- **Geometry (space)**: π, √2, √3, the golden ratio φ, the dihedral angle of the regular tetrahedron arccos(1/3) ≈ 70.53° (five of them fall short of 360° by 7.356°), the complement of the magic angle 35.264°, the solid angle 4π.
- **Evolution (phase, time)**: e (the exponential map of a group), 2π/n (one step of U^n = I), e^{iπ} = −1 (connecting geometry and evolution), rapidity.
- **Statistics (an easily overlooked system)**: √(2π), Euler's constant, ln 2, ζ(3), $π^4/15$, the constants of Wien's displacement law, Onsager's critical temperature 2/ln(1 + √2), percolation thresholds, the Feigenbaum constants, the dimension 0.868 of the devil's staircase, the golden-mean critical constant of the circle map, Khinchin's constant, Lévy's constant $π^2/(12 \ln 2)$, the critical coupling of the Kuramoto model, critical exponents.

9.4 Derived system (numbers that look like magic numbers but are derived from other numbers)

$α^{-1}$ = 137.035999… (the electromagnetic coupling, running with energy), the Thomson cross-section (8π/3) r_e^2, the first term of the anomalous magnetic moment α/2π, fine-structure splitting ∝ $α^2$, the Rydberg constant $α^2$ m_e c/(2h), the Bohr radius $ƛ_C/α$, $\sin^2θ_W$, the proton–electron mass ratio, the strong coupling, the gravitational coupling. These should be assembled by the readout map from numbers of the integer and irrational systems, and they appear as cross-sections and interaction coefficients.

9.5 Pairs that are near but not equal

Structural number	Observed or nearby number	Explanation	Status
137	$α^{-1}$ = 137.036	Unexplained. Candidate for locking and quasi-oscillation among several structural numbers	Candidate
640320^3 + 744	e^{π√163} (smaller by 7.5 × 10^{-13})	Explained by class number 1 and the expansion of the j-function [X32]	Established
196883 + 1	The coefficient 196884 of the j-function	The nearness revealed a hidden symmetry (the Monster) and became a theorem [X31]	Established
2/3	Koide's value	Unexplained	Candidate
Lattice orders 4 and 6	Order of the exchange	Not crystallographic; locking in the fourteenth [S19]	Theory

e^{π√163} and moonshine are examples where "near but not equal" comes from the

structure of number theory and group theory itself, and they are precedents for a method of finding hidden symmetries from near coincidences.

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10. Confirmed Results

The checks in this section were made only with the mathematics of objects already in the theory, without using the readout map (which is open). No parts were added. Programs and results are in the folder of Appendix A.

10.1 Check 1: true square and norm

Proposition. A linear map T that preserves both the sum of true squares $z^{T}z$ and the norm $z^{\dagger}z$ is necessarily a real orthogonal map.

Proof. From $T^{T}T = I$ and $T^{\dagger}T = I$ we get $T^{T} = T^{\dagger}$, that is, $\bar{T} = T$, so T is real. Being real with $T^{T}T = I$, it is orthogonal. For generators, the intersection of complex antisymmetric and anti-Hermitian matrices is the real antisymmetric matrices. □

Numerically, in eight dimensions, complex orthogonal maps preserved the true square (change 3.6×10^{-14}) and changed the norm (change 100), unitary maps preserved the norm (1.8×10^{-15}) and changed the true square (2.9), and real orthogonal maps preserved both (10^{-15}). The dimension of the intersection of generators was 28, equal to the dimension of real antisymmetric matrices in eight dimensions.

Relocating terms. With D = diag(1, …, i, …), which multiplies coordinates by i, an element Λ of the group O(p, q) of signature (4,0), (3,1) or (2,2) is mapped to $D\Lambda D^{-1}$, and all of these became elements of the same $O(4, \mathbb{C})$ (error 10^{-16}). Exchanging space and time corresponds to a different real slice of the same complex quadric.

10.2 Check 2: triality of D_4

What was checked	Result
Automorphisms of the D_4 root system	1152 (the order of $W(F_4)$)
Action on the three cosets of D_4^*/D_4: v (odd class), s and c (two half-integer classes)	Each of the 6 permutations occurs 192 times; triality is present in full as S_3
An element T of order 3 cycling v → s → c	Exists
The 24 Hurwitz units	Split into 8 each in v, s, c; T cycles the three octets and preserves the set of 24
The pair (T, T)	Preserves the 240 roots of E_8
The centre (½,½,½,½) of the ball in which 137 was counted	In class s; T sends it to class c, T^2 to class v
Quaternion form of T	$\mathbf{q} \mapsto \mathbf{a}\,\mathbf{q}\,\mathbf{a}$, a = (−1 + i + j + k)/2 (a unit of order 3); the operation of multiplying by the same unit from left and right

The 24 units of the equally spaced gauge of the R = 3 system split into one vector set and two spinor sets, and triality exchanges them by left and right multiplication. The subgroup fixed by triality is G_2, and G_2 contains SU(3) (a mathematical fact). Whether this corresponds to physical colour or to three generations is a candidate.

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11. Correspondence with Published Papers

The 156 published papers (13 series) were mapped to the rows of the catalogue. 81 rows have a published standard; the 15 rows to which no paper corresponds are W4, K3, K6, K7,

S13, S15, E4, E5, G12, C3, C4, L4, L5, L8 and L9. These 15 rows are supported by external literature or by the checks of this paper. The full mapping table is in Appendix B.

11.1 Resetting the axiom numbering

Document	Old number	Number in this paper
Basic axiom system v9.1 [S1]	Axiom 0 anonymity	A1
Same	Axiom 0.5 scale anonymity	Consequence of A3, S20
Same	Axiom 0.6 dimension-wise central-projection readout	No corresponding row (a readout rule for curvature correction)
Same	Axiom 0.7 existence of anonymous partial projection	P1
Same	Axiom 1 all-positive-sign zero closure	A3 (moved from axiom to theorem)
Same	Axiom 2 nontrivial existence	Premise of A3
Same	Axiom 3 degeneracy of the projection axis and imaginary-direction readout	G17
Same	Axiom 15 discreteness of observed phase	A4, G13
Zero closure, finite order, self-consistent geometry [S3]	A1 complex zero closure	A3
Same	A2 finite order	A4
Same	A3 simplex consistency	No corresponding row (later found to impose no constraint [S12])
Same	A4 self-consistency	A2
Earlier N-body series	Axiom 1, first axiom (zero closure)	A3
Research project design [S14]	First-principle candidate (anonymous distinguishable states)	A1
First and second thought experiments [S15, S16]	P1 anonymity, A1 a conserved readout	A1, A3 and K1
Dual-geometry series	$\nu\lambda = 1$, zero-point ½, one asymmetric bit	The conjugate pairs of §4.2, the integer ratio ½, S12

11.2 Revision of a claim

Papers 7 and 9 and the supplement of the BH thermodynamics programme [S29, S30] took the locking side, filling the gap between 137 and α^{-1} with a closed identity. This paper revises that to the position that reads the gap itself as content (§4.3, §9.5). The earlier papers are cited as the record of the position before revision.

11.3 Interpretations of numbers kept as open problems

For the following disagreements, no single interpretation is taken as correct. That several interpretations are possible is the present answer, and these remain as problems.

Item	Interpretations
Rotation orders of the four-dimensional lattice	Crystallographic 1, 2, 3, 4, 6 (§3.4) / 1, 2, 3, 4, 5, 6, 8, 10, 12 of §20.3 of the unified hypothesis paper [S21]
Decomposition of 137	1 + 8 + 128 [S19] / 1 + 8 + 24 + 40 + 64 [S21]
Oddness	Odd R (G10) / a single state has odd R^2 (paper 5 of the dual-geometry series)
Number of components per generation	16 (G15, E4) / 15 [S4]
Order of the exchange	248 (clock of observation) / 124 (order of the root) [S7]. The lock-dynamics paper finds that locking does not select order 124 [S26]
Parity and statistics	Type of statistics (G14, S17) / a conventional label
Cross term 2ab	Mass (S9) / oscillation of curvature (the complex-rotating-wave paper)
Time reversal and the arrow of time	One side (S12, W1) / the forced bit is separate from the arrow of time (paper 14 of the dual-geometry series)
Axis Q	Three axes $Q_1Q_2Q_3$ and 16 states (G15) / one axis with 8 values and 62 states (W8, the three-layer model)
Lorentz signature	Assumed (gnomonic 1) / derived (gnomonic 10, S3)
How SU(3) is obtained	Through G_2 (S8) / by the 3 + 2 decomposition [S18]

11.4 Withdrawn or corrected claims

When citing, the corrected version is the standard. “Zero Closure Was Four-Dimensional” v4 withdrew “compactness implies a discrete spectrum” [S5]. Three-layer model 7 and D1 v2.1 withdrew H_xyz_inphase and δδt. Paper 5 of the BH thermodynamics programme corrected the “disagreement” in the abstract of paper 1 as a confusion between 3 + 1 and 4 + 1. Paper 8 of the BH thermodynamics programme changed its title to “structural correspondence” in v2. Papers 6 and 12 of the dual-geometry series and the observer–system beat conjecture have also been corrected.

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12. Position among Prior Studies

There are prior studies close to individual elements.

- **Space from Hilbert space**: Cao, Carroll and Michalakis [X1] assumed a tensor-product decomposition into subsystems and assembled space from mutual information. Carroll and Singh [X2] took only the list of eigenvalues of the Hamiltonian and the components of one vector as the minimal ontology. Both give time evolution from outside and place the tensor-product decomposition (labels of subsystems) first; from the side of anonymity, the latter is a hidden assumption. Stoica [X3] criticized that three-dimensional space does not emerge uniquely from the minimal quantum structure and that a choice is needed. This points to the same place as this paper’s “until the projection method is chosen, what is readable is undetermined” (§4.4).
- **Philosophy of symmetry**: from the classics of Weyl [X22] and Wigner to the collection on the philosophy of symmetry in physics [X21].
- **Anonymity**: permutation invariance is considered a postulate consistent with quantum theory but not derivable from it [X4]. Structural realism holds that what exists in the world is only the structure of relations [X5, X6]. A Japanese exposition of relationalism (Leibniz, Mach), which regards spacetime as relations among things, is [X33].

- **Self-consistency**: Chew's bootstrap combined self-consistency with the absence of elementary particles ("democracy") but placed spacetime as a background [X7].
- **A world of ratios only**: Barbour's shape dynamics holds that only ratios carry objective information and showed that imposing scale invariance makes the energy and angular momentum of an island universe zero [X10, X11]. This has the same form as the closure of this paper but does not treat discreteness or magic numbers.
- **Dimensionless numbers**: Duff holds that the number of dimensional fundamental constants is zero and that only dimensionless numbers are fundamental [X13].
- **Noether's theorem**: the second theorem and strong laws [X8], quantum anomalies [X9], the view of symmetries as meta-laws [X14].
- **Duality**: the schema of De Haro and Butterfield [X12]; Kaluza–Klein momentum and charge [X15].
- **Triality and three generations**: the proposals of Boyle and of Furey and Hughes start from octonions and Clifford algebras and are at the stage of conjecture [X16, X17].
- **Attempts to derive α from structure**: Eddington, Rosen, Smith and claims from E_8 all lie on the side demanding agreement [X18, X19, X20].
- **Locking**: the devil's staircase of the circle map and the renormalization of the Farey tree [X23, X24]; the modular group in the quantum Hall effect [X25].
- **Precedents for magic numbers**: nuclear magic numbers [X30], moonshine [X31], e^{π√163} [X32].

Within the range examined, the following combinations were not found: (1) a construction taking anonymity and self-consistency as axioms and obtaining closure and return as two theorems; (2) an axiom that what is attached to a symmetry is not a conserved quantity but a dimensionless number; (3) an argument combining conjugacy and duality with anonymity to merge separate symmetries as different views of one symmetry; (4) the position reading "near but not equal" as locking and quasi-oscillation among several structural numbers; (5) a route into three generations from the triality q ↦ a q a built into the Hurwitz lattice. Details of the survey are in Appendix C.

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13. Open Problems

13.1 The root difficulty

What is visible depends on the projection method (§4.4). The map from the state space $\{z_n\}$ to spacetime {xyzt} must be at least a partial projection. The projection method is unknown.

13.2 The readout map

One z_n is merely a point in a two-dimensional plane. How do three independent spatial directions and one time assemble from the relations among many z_n? It is not known how a set of complex numbers carries the quaternion structure required by the condition on composition algebras. Standard theory proceeded without this map: it accepted discreteness as a result, gave the background spacetime and the ruler by hand, and advanced with the computational methods of bra–ket notation, gauge theory and renormalization. Weyl made R carry the role of Q in 1918 and discarded R from gauge theory in 1929.

13.3 The unit of the ruler and the two orders

The closure does not fix the ruler (§3.3). What fixes the unit of R? How do the order of the exchange and the lattice orders 4 and 6 combine? These two are different faces of the fact that the readout map is unknown. Candidates are the conjugacy of amplitude and phase (§4.2) and the locking of the projection ratio (§4.4).

13.4 Discipline of numerical experiments

The catalogue of this paper is also the pass/fail standard for numerical experiments. The author's model built on the 600-cell imported a Dirac-type generator, derivatives of position, spin matrices and a Poisson equation, none of which are in the skeleton, and passed all stages by swapping parts and readings each time a result did not match. That pass is not a derivation from the skeleton. Later numerical experiments are to be built only from the finite operations in the skeleton (the Clifford translation multiplying a unit quaternion from the left and the array of component rotations multiplied from the right); as soon as a part not in the catalogue turns out to be needed, the work stops and the part is stated explicitly as a new assumption. The group 2I of the 600-cell contains five-fold and ten-fold rotations and lies outside the crystallographic restriction, so it does not satisfy the condition of discreteness of spacetime (§3.4).

13.5 Checking the consequences of the distinguishability axiom

The following have not been checked for the distinguishability axiom (A1′). (1) If the indistinguishability of layers is carried through, does it require the same definition to hold at the layers below the square of z' (z^4 and beyond)? If so, power sums vanish one after another and may connect to the return $U^n = I$. (2) In the exponential representation $E = \ln|z| + i\varphi$, $z \mapsto z^2$ is $E \mapsto 2E$, a discrete scale symmetry with factor 2. It is a semigroup symmetry with no inverse, outside Noether's theorem. (3) At the squared layer z and $-z$ cannot be distinguished, and the information of parity (sign reversal at 360°) disappears. The disagreement between the exchange orders 248 and 124 (§11.3) is a candidate for the difference in reading the same rotation at two layers. (4) The relation between the choice of layer of the squaring map and the four readings by the combination of t and R ($xyz + t$, $Q_1Q_2Q_3 + R$, $xyz + R$, $Q_1Q_2Q_3 + t$) (Appendix D).

13.6 Checking the correspondences listed as candidates

Whether the groups K1–K7 agree up to renaming alone. Whether the duality of inside and outside means the exchange of particle and absorber. Whether triality connects to three generations and to the relation between bosons and fermions. The correspondence between the 16 sign vectors and the spinor 16 of SO(10). Agreement between the Hopf winding number and the charge of sign vectors. The interpretations of numbers in §11.3.

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14. Conclusion

This paper fixed, as a catalogue with statuses and provenances, the symmetries and magic numbers that a research programme placing no background spacetime must satisfy. The zeroth axiom was anonymity and the first axiom self-consistency; the distinguishability axiom was derived from anonymity; the single definition of closure was the linear zero closure; the square-sum closure was shown to be the same definition read at the squared layer; and the return of time evolution was a theorem. Below them we placed big-picture symmetries, conjugacy and duality, the dimensionless numbers attached to symmetries and the projection symmetry, and listed the Standard Model, general relativity, E_8, the algebra and geometry of this system and the locking symmetries in the same format.

Mathematically, we confirmed that only real orthogonal maps preserve both the true square and the norm, and that the 24 Hurwitz units are cycled 8 + 8 + 8 by the triality $q \mapsto a\,q\,a$, which preserves the roots of E_8. We mapped the 156 published papers to the catalogue, reset the axiom numbering, revised the earlier claim on the gap of α, and kept the numerical disagreements as problems admitting several interpretations.

The root difficulty of the map between state space and spacetime lies in the dependence of what is visible on the projection method. Since spacetime is stably readable, the map must be at least a partial projection. In the light of one of the success conditions of the research project design [S14], "to show that certain requirements are mathematically incompatible",

this paper lays out each requirement and its provenance in one table and shows which requirements are already confirmed, which disagree with each other and which are open. Later papers and numerical experiments take the numbering and statuses of this paper as their answer-check standard.

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Appendix A: Reproducing the Checks

The programs and results of Checks 1, 2 and 2b are in the repository folder `次元の生成構造/四次元球殻住民_自己無撞着_視野角三角形_20261016/見落とされた対称性の確認_20261017/`. They are reproduced by `./run_all.sh`, and the results are written as JSON to `results/`.

Appendix B: Self-Citation Correspondence Table

The correspondence between the 156 published papers and the catalogue rows, the disagreements and the correspondence with the old numbering are in `自己引用の対応表_20261017.md` (self-citation correspondence table, in the same folder as this paper; in Japanese).

Appendix C: The Original Lists and the Prior-Study Survey

The working documents on which this paper is based are (in Japanese):

- `次元の生成構造/対称性の一覧_保存量と固定点_20261017.md` (list of symmetries)
- `次元の生成構造/マジックナンバーの一覧_20261017.md` (list of magic numbers)
- `次元の生成構造/対称性の一覧_先行研究調査_20261017.md` (survey of prior studies)

Appendix D: Dialogue Record of the Thought Experiment on Linear Zero Closure

The distinguishability axiom and the definition of linear closure in §3.0 came from the dialogue between the author and Claude on 2026-10-18. The dialogue record is in `次元の生成構造/線形和のゼロ閉塞_思考実験_20261018/線形和のゼロ閉塞 Σzn=0 仮定の追加の思考実験_20261018.md` (no summarization; only typos in the author's remarks corrected; in Japanese).

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